

Tutorial-10

Sn	Age	Income	Credit Score	Loan Approved
S ₁	25	40	650	Yes
S ₂	30	60	700	Yes
S ₃	45	35	600	No
S ₄	50	30	580	No
S ₅	28	55	720	Yes

Target Encoding

Yes = 1 ; No = 0

Questions:

- 1) Assign initial wts to all 5 Samples
- 2) Construct a Decision Stump using feature 1 (Age) to classify the Samples
- 3) Identify misclassified Samples & Compute & compute the weighted error rate
- 4) Calculate classifier wt.
- 5) Update Sample wts for next iteration.

Solns

- AdaBoosting (Adaptive Boosting)

- 1) Initialise wts for all
- 2) Check how many correct classification and in correct classification for each attribute
- 3) Find weighted error.

$$E_{w2} = (\text{no. of incorrect ones}) \times (\text{weights of each})$$

4) $\alpha_x = \frac{1}{2} \log_e \left(\frac{1 - e_x}{e_x} \right) \rightarrow$ performance of Decision stump

- 5) Calculate new wts for each

$$w_{\text{new}} = w_{\text{old}} * e^{\alpha_1} \rightarrow \text{Incorrect classification}$$

$$w_{\text{new}} = w_{\text{old}} * e^{-\alpha_1} \rightarrow \text{Correct classification}$$

- 6) Normalise weights $\rightarrow \frac{w_{\text{new}}}{(\text{Total})} \rightarrow$ By adding all wts

- 7) For all attributes follow same

- 8) Final Predicted Values

$$P = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \dots + \alpha_n h_n(x)$$

$$h_1(x) \rightarrow \text{Yes/No label}$$

$$h_1(x) \rightarrow \text{No} \rightarrow \text{label}$$

If $P > 0 \rightarrow \text{Yes}$ } After calculating
 $P < 0 \rightarrow \text{No}$ } After average.

Adaptive Boosting is an example ensemble machine learning that combines multiple weak learners (called decision stumps).

It forms a strong predictive model.

Step-1: Given Data.

Sample	Age	Loan	Approved
S_1	25	Yes	+1
S_2	30	Yes	+1
S_3	45	No	-1
S_4	50	No	-1
S_5	28	Yes	+1

Assign initial weights.

Total Samples = 5

Weights (w_i) = $\frac{1}{5} = 0.2$

$S_1 = S_2 = S_3 = S_4 = S_5 = 0.2$

Step 2. Construct Decision Stamp.

(Feature Vs Age)

Sort Values

Sample — Age — loan Approved

S ₁	25	1
S ₅	28	1
S ₂	30	1
S ₃	45	-1
S ₄	50	-1

Decision Threshold: $\text{Age} \leq 28 \rightarrow \text{Yes}$

$\geq 28 \rightarrow \text{No}$

Step 3:

Missclassified Samples are S₂

Where table is:

Sample	Age	loan Approved	predicted
S ₁	25	1	
S ₅	28	1	
S ₂	30	1	
S ₃	45	-1	-1
S ₄	50	-1	-1

Step 4: Wt error

$$e = 1 \times (0.2) = 0.2$$

Step 5:

Calculate α (Needed for weight update)

$$\alpha = \frac{1}{2} \log \left(\frac{1-e}{e} \right)$$

$$= \frac{1}{2} \log \left(\frac{1-(0.2)}{0.2} \right)$$

$$\boxed{\alpha = 0.693}$$

Step 6: Updated wts.

For correctly classified $e = e^{-\alpha} = e^{-0.693} = 0.5$

Wrongly classified $e = e^{\alpha} = e^{0.693} = 2$

Previous weight

$$S_1 = S_2 = S_3 = S_4 = S_5 = 0.2$$

New wts. $S_1 = S_3 = S_4 = S_5 = 0.5 \Rightarrow 0.2 \times 0.5 = 0.1$

$$S_2 = 2 \Rightarrow 0.2 \times 2 = 0.4$$

$$S_1 = 0.1, S_2 = 0.4, S_3 = 0.1, S_4 = 0.1, S_5 = 0.1$$

$$\text{Total Wt} = 0.1 + 0.4 + 0.1 + 0.1 + 0.1 = 0.8$$

Normalise wts

Normalise wts

Sample

Final wts

S₁

$$0.168 = 0.125$$

S₂

$$0.4108 = 0.5$$

S₃

$$0.1108 = 0.125$$

S₄

$$0.1108 = 0.125$$

S₅

$$0.1108 = 0.125$$